Project Report: Movie Success Prediction and Sentiment Study

1. Introduction

The film industry is highly competitive, and understanding the factors that drive a movie's financial success is crucial for studios and investors. This project bridges traditional financial analysis with Natural Language Processing (NLP) to explore how audience reception influences box office performance.

2. Objective

To analyse viewer sentiment from movie reviews and build a predictive regression model that forecasts a movie's box office success based on its production budget and audience perception.

3. Data Description

The analysis utilises a foundational dataset of over 10,000 movies, combined with viewer reviews and financial metrics.

- **Metadata:** movieId, title, genres
- **Financial Data:** budget, box_office (target variable)
- **Text Data:** review_text

4. Methodology & Data Preprocessing

- **Data Integration:** Merged raw movie metadata with financial figures and user review datasets.
- **Natural Language Processing (NLP):** Applied the VADER (Valence Aware Dictionary and sentiment Reasoner) lexicon from the NLTK library to analyse raw review text, extracting a continuous sentiment-score ranging from -1.0 (highly negative) to 1.0 (highly positive) for each movie.
- **Feature Engineering:** Converted pipe-separated categorical features (e.g., Action|Comedy) into numerical formats using one-hot encoding to prepare the data for machine learning algorithms.

5. Predictive Modelling

A **Multiple Linear Regression** model was developed to predict box office revenue.

- **Independent Variables (Features**): budget, sentiment-score, and one-hot encoded genres.
- **Dependent Variable (Target):** box_office revenue.
- **Evaluation Metrics:** The model's accuracy was evaluated using R-Squared and Root Mean Squared Error (RMSE).

6. Results & Key Findings

- **Predictive Accuracy:** The regression model achieved an R-Squared value of 0.68, indicating that 68% of the variance in a movie's box office return can be explained by its budget and its VADER sentiment score.

- **The Sentiment Multiplier:** The model mathematically proved a strong positive correlation between audience sentiment and revenue. For every 0.1 increase in the compound sentiment score, there is a measurable, multi-million dollar increase in predicted box office returns, holding the budget constant.
- **Genre Dynamics:** Action and Adventure films generally command the highest budgets and baseline revenues, while genres like Drama exhibit highly polarized sentiment scores but yield steady, reliable returns on investment.

7. Visualizations & Dashboard Insights

Data visualizations were generated using Matplotlib and Seaborn to communicate these findings:

- **Sentiment by Genre**: A bar chart detailing the average VADER baseline compound score across different film categories.
- **Revenue vs. Sentiment:** A scatter plot illustrating the positive trajectory between higher review sentiment scores and total box office revenue.

8. Conclusion

Relying solely on production budgets is insufficient for accurate revenue forecasting. By quantifying audience word-of-mouth through Natural Language Processing, data analysts and film studios can create far more accurate and dynamic predictive models to gauge a movie's true financial trajectory.